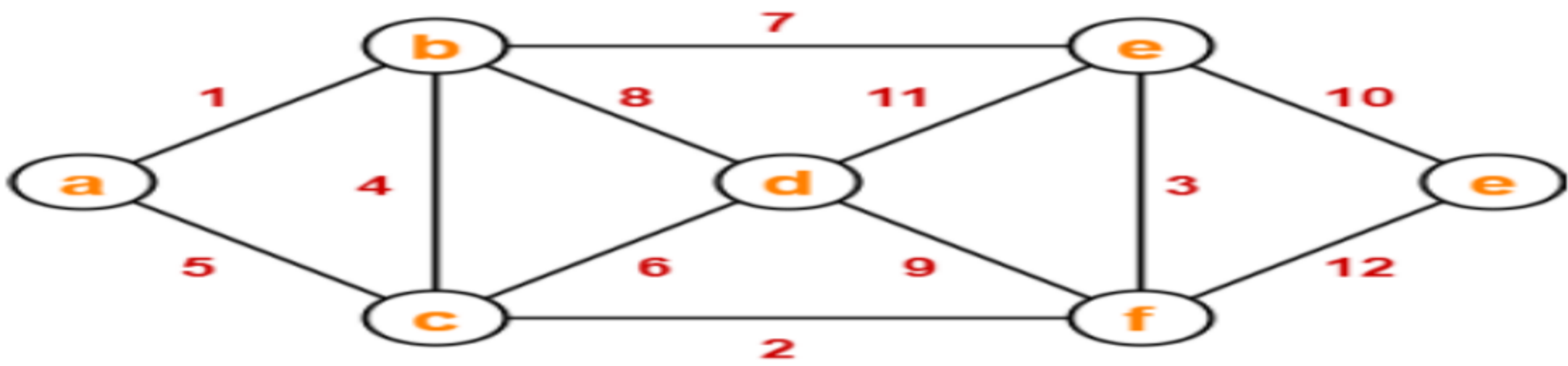
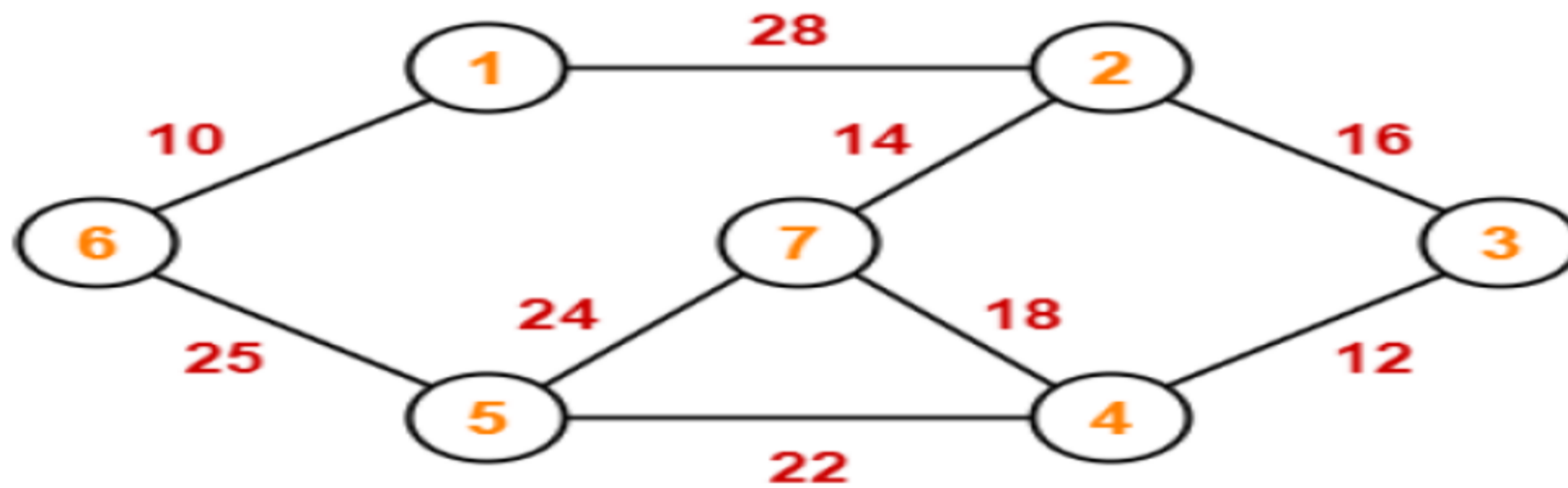


SEMESTER EXAMINATION-2022	
CLASS - III rd YEAR SUBJECT:DESIGN AND ANALYSIS OF ALGORITHM	
PAPER CODE: BCE-C513	
Time: 3 hour	Max. Marks: 70 Min. Pass: 40%
Note: Question Paper is divided into two sections: A and B . Attempt both the sections as per given instructions.	
SECTION-A (SHORT ANSWER TYPE QUESTIONS)	
Instructions: Answer any five questions in about 150 words each. Each question carries six marks.	(5 X 6 = 30 Marks)
Question-1: What is time and space complexity of an algorithm?	
Question-2: Prove the assertion $\frac{1}{2} n(n-1) \in \Theta(n^2)$.	
Question-3: Prove the assertion $100n+5 \in \Theta(n)$.	
Question-4: Explain recurrence equation.	
Question-5: Explain hierarchy theorems.	
Question-6: Solve the following recurrence relation using Master's theorem. $T(n) = 3T(n/2) + n^2$	
Question-7: Explain convex hull.	
Question-8: Explain greedy method and its applications.	
Question-9: Find the complexity of given recurrence: $T(n) = \begin{cases} 3T(n-1), & \text{if } n > 0 \\ 1, & \text{otherwise} \end{cases}$	
Question-10: Solve the following recurrence relation using Master's theorem. $T(n) = 2T(n/2) + n \log n$	
SECTION-B (LONG ANSWER TYPE QUESTIONS)	
Instructions: Answer any FOUR questions in detail. Each question carries 10 marks.	(4 X 10 = 40 Marks)
Question-11: Construct the Minimum Spanning Tree (MST) for the given graph using prime's algorithm.	
	
Question-12: Explain backtracking in detail.	

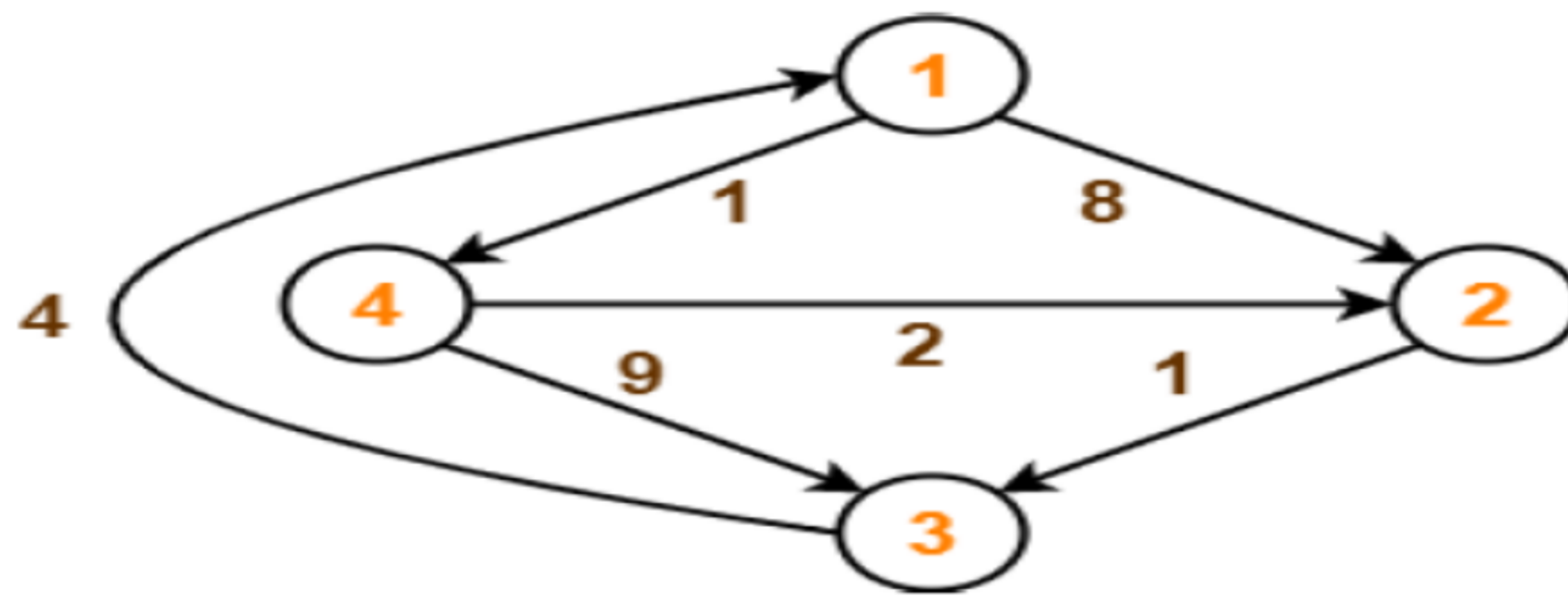
Question-13: Construct the Minimum Spanning Tree (MST) for the given graph using Kruskal's algorithm.



Question-14: Explain travelling salesman problem in detail by drawing a neat figure.

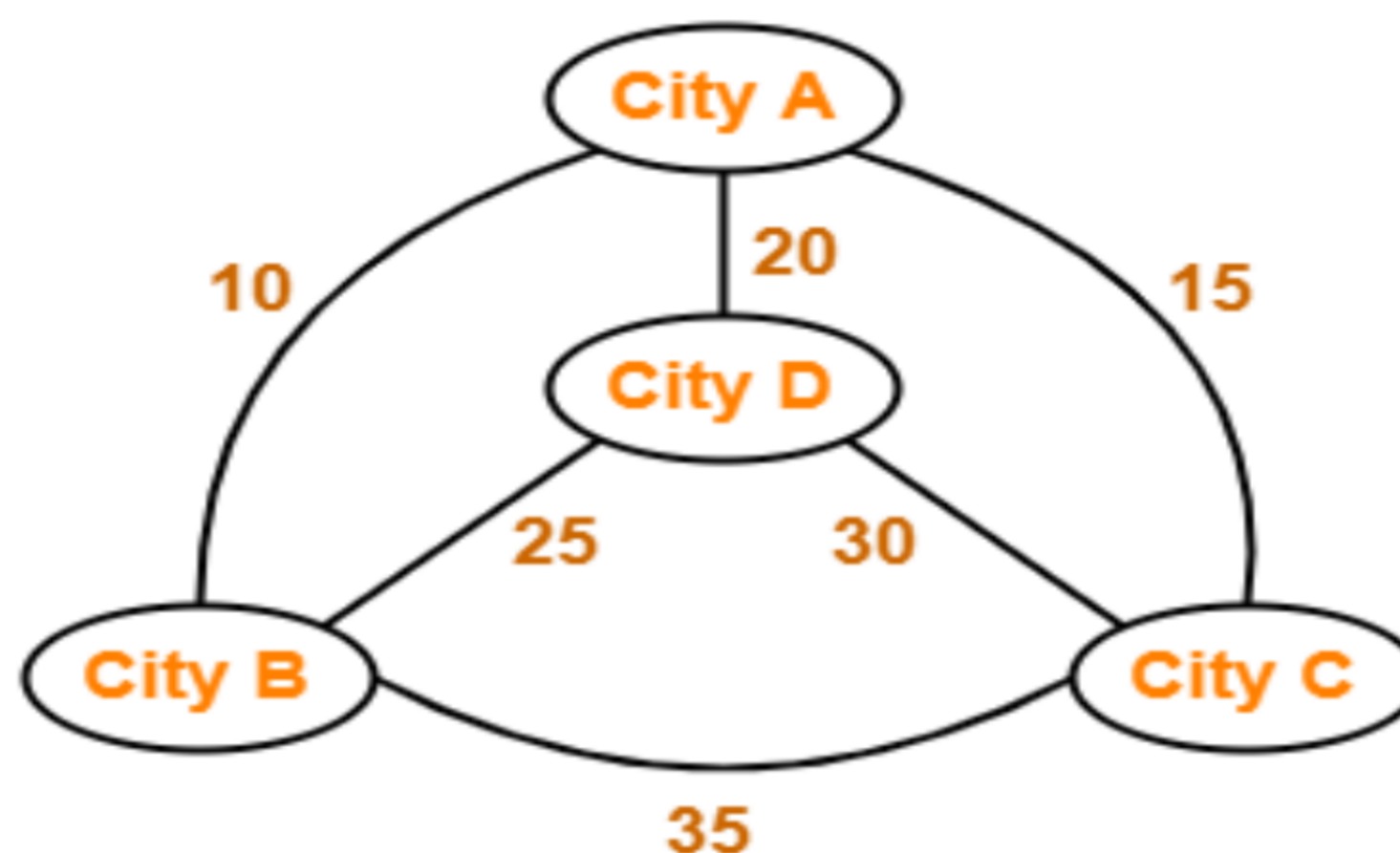
Question-15: Explain divide and conquer technique with the help of a suitable diagram.

Question-16: Using Floyd and Warshal algorithm find the shortest path distance between each pair of vertices of the diagram given below:



Question-17: Explain Dijkstra's algorithm with the help of diagrams.

Question-18: The following graph shows a set of cities and distance between every pair of cities-



If starting city is A, then find the TSP tour of the above graph.